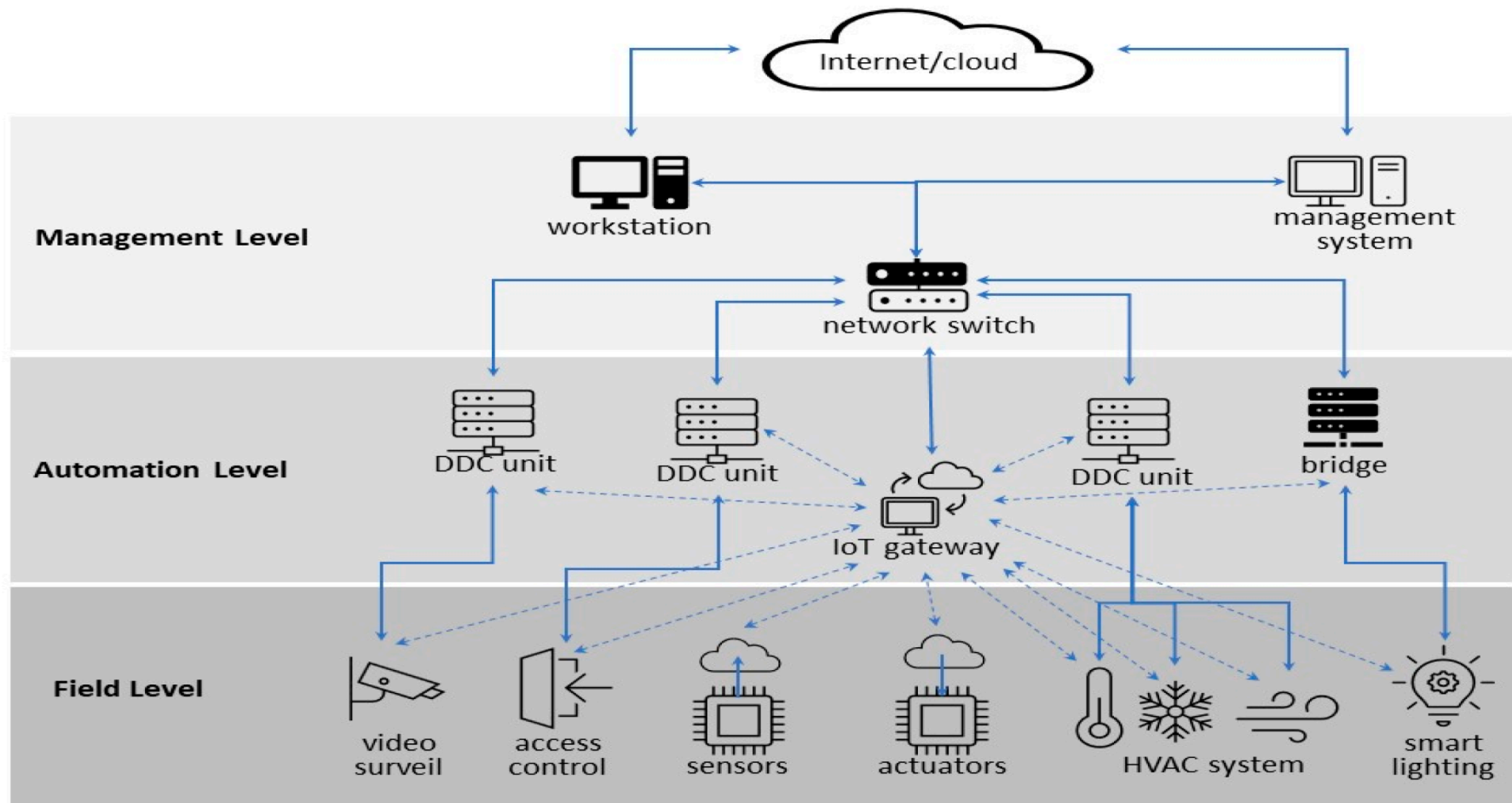


Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	31 January 3035
Team ID	
Project Name	PocketSmart AI (Budget & Recommendation Assistant)
Maximum Marks	4 Marks

Technical Architecture:



Reference:

<https://docs.streamlit.io/>

Technical Architecture: PocketSmart AI

Table-1 : Components & Technologies

S.No	Component	Description	Technology
1.	Frontend Interface	User dashboard for scenario inputs, constraint setup, and financial ledger display	Streamlit (Python Web Framework)
2.	Application Logic-1	Authentication handling, API key setup, and global state initialization	Python (<code>st.session_state</code>)
3.	Application Logic-2	Dynamic system and scenario prompt construction for home decor, events, and jewelry	Custom Python Orchestrator
4.	Application Logic-3	Transaction aggregation, budget metric calculations (Income, Expenses, Balance)	Pandas DataFrames

5.	Data Storage / In-Memory	Session persistence layer for transaction history and user context	Streamlit Session State (<code>st.session_state</code>)
6.	External API / AI Model	Generative AI engine for cross-platform itemized shopping recommendations	Google Gemini 3.8 Flash API (<code>gemini-3.8-flash</code>)
7.	SDK Client	Integration library for secure communication with Google Cloud Generative AI	<code>google-genai</code> Python SDK
8.	Visualization Engine	Interactive graphical rendering of expense category breakdowns	Plotly Express Engine
9.	Cloud Infrastructure	Hosting, execution, and public deployment platform	Streamlit Community Cloud
10.	Source Control	Version control, codebase management, and continuous delivery pipeline	GitHub

Table-2:Infrastructure & Component Mapping :

S.No	Category	Component Name	Execution Environment	Interface / Dependency
1.	Client / User Interface	Desktop & Mobile Web Browser	Client Side (User Device)	HTTPS / Web Engine
2.	Frontend Platform	Streamlit App Interface	Cloud Hosted / Server Side	Python Runtime (v3.9+)
3.	Backend Application Logic	Prompt Engine & Ledger Services	Cloud Hosted / Server Side	Pandas, Session State
4.	Third-Party AI API	Google Gemini 3.8 Flash API	Cloud Managed (Google Cloud)	REST / <code>google-genai</code> SDK

5.	Storage Component	In-Memory Session Storage	Temporary / Volatile Memory	st.session_state
6.	Analytics & Charts	Plotly Express Visualizer	Server Side rendering to Web UI	Plotly JavaScript Core

References:

[Streamlit Framework Document](https://docs.streamlit.io/):

<https://docs.streamlit.io/>

[Google Gen AI Python SDK Repository:](https://github.com/googleapis/python-genai)

<https://github.com/googleapis/python-genai>

[Pandas Data Analysis Library Documentation:](https://pandas.pydata.org/docs/)

<https://pandas.pydata.org/docs/>

[Plotly Express Python Visualization Documentation:](https://plotly.com/python/plotly-express/)

<https://plotly.com/python/plotly-express/>